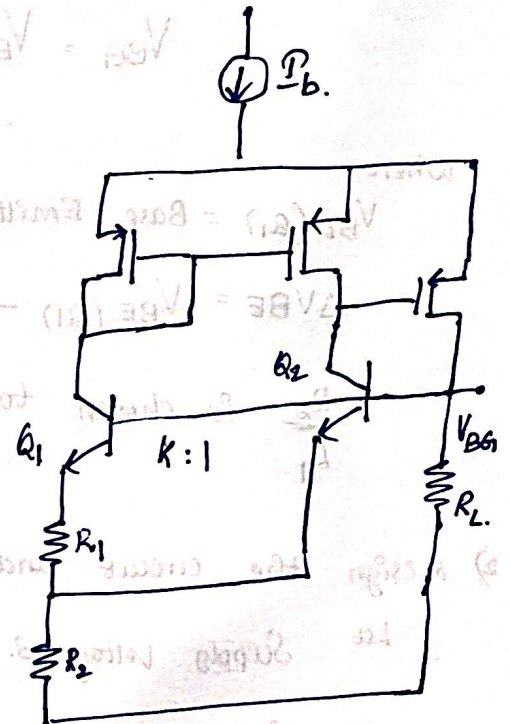
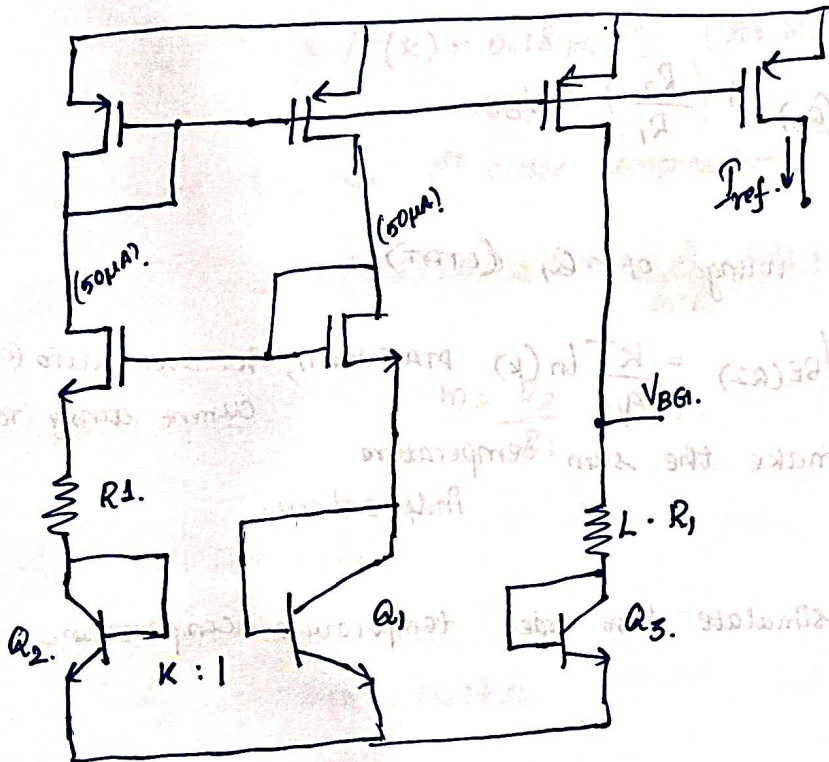


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PRE: LAB : 7 : Current Reference and Bandgap Circuits :-

To design and Simulate different bandgap circuits :-



Solution:-

$$V_{BG1} = V_{BE1} + k \Delta V_{BE}.$$

$$\Delta V_{BE} = V_{BE1} - V_{BE2} = \frac{kT}{q} \ln\left(\frac{J_1}{J_2}\right)$$

Where J_1/J_2 is the current density ratio (8:1 here)

$$V_{BG1} = V_{BE1} + k \cdot \frac{kT}{q} \ln(8).$$

Circuit-1

1. Deriving the expression for V_{BG} :-

The bandgap voltage V_{BG} is obtained summing a PTAT and CTAT (Complementary to absolute temperature, typically the base-emitter voltage of a BJT, V_{BE})

Expression:-

$$V_{BG} = V_{BE}(Q_1) + \left(\frac{R_2}{R_1}\right) \Delta V_{BE}$$

Where:-

$V_{BE}(Q_1)$ = Base Emitter voltage of Q_1 (CTAT)

$$\Delta V_{BE} = V_{BE}(Q_1) - V_{BE}(Q_2) = \frac{kT}{q} \ln(k) \quad \text{PTAT term, } k = \text{area ratio (or) current density ratio}$$

$\frac{R_2}{R_1}$ is chosen to make the sum temperature Independent.

2) Design the circuit and simulate first order temperature compensation.

Let Supply Voltage = 3.

Current Density Ratio = $k = 8$.

Nominal Emitter area for Q_1 = A

Emitter area for Q_2 = $8A$

Base Current in Q_1 and Q_2 $I_{BQ} = 50 \mu A$.

$V_{BE1} \approx 0.7V$ at $27^\circ C$.

Step:1

$$\Delta V_{BE} = \frac{kT}{q} \ln(k)$$

$$\frac{kT}{q} = 0.02585$$

$$\Delta V_{BE} = 0.02585 \times 2.08 = 0.0538V$$

Set PTAT/CTAT Balance:

$$\frac{dV_{BG}}{dT} = \frac{dV_{BE}}{dT} + m \frac{d(\Delta V_{BE})}{dT}$$

$$\frac{dV_{BE}}{dT} = -2 \text{ mV}/^\circ\text{C}$$

$$\frac{d(\Delta V_{BE})}{dT} = \frac{k}{q} \ln(k) \approx 0.18 \text{ mV}/^\circ\text{C} \quad (\text{for } k=8)$$

Set Sum = 0, for 1st order compensation.

$$-2 + m(0.18) = 0 \Rightarrow m = \frac{2}{0.18} \approx 11.1$$

$$m = \frac{R_2}{R_1} \approx 11.1$$

Calculate resistor value,

$$R_1 = 1 \text{ k}\Omega$$

$$R_2 = 11.1 \times 1 \text{ k} = 11.1 \text{ k}\Omega$$

$$V_{BG} = V_{BE1} + m \Delta V_{BE}$$

$$= 0.7 + 11.1 \times (0.0538) = 1.3 \text{ V}$$

3) For start-up Circuit:-

The start-up circuit uses an NMOS device whose gate is connected to the bandgap output (V_{BG}), source to ground, and drain to the current mirror bias node. It is sized to provide sufficient current @ power-on by using the equation,

$$I_{\text{Startup}} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{DD} - V_{th})^2$$

and to turn-off, once the bandgap output reaches its normal level, the aspect ratio W/L is chosen to achieve this condition.

Circuit: 2 :-

Expression for V_{BG} (derive expression)

Currents through Q_1 and Q_2 ,

ΔV_{BE} calculation:-

$$\Delta V_{BE} = V_{BE1} - V_{BE2} = \frac{kT}{q} \ln\left(\frac{I_1}{I_2}\right) = \frac{kT}{q} \ln k.$$

$k = \frac{\text{current density of } Q_1}{\text{current density of } Q_2} = 8$ for my specs;

Total Output,

The output node @ V_{BG} is the typical bandgap weighted sum:-

$$V_{BG} = V_{BE1} + \alpha \Delta V_{BE}.$$

Where α (the weighting multiply for ΔV_{BE}) is controlled by resistor division. From Widlar structure:

$$V_{BG} = V_{BE1} + \frac{R_2}{R_1} \Delta V_{BE}.$$

5) Design for First-Order Temperature Compensation and Current Reference,

A) First Order Temp Comp: Find Needed $\frac{R_2}{R_1}$ Ratio,

The CTAT (V_{BE} drops with T) and PTAT (V_{BE} rises with T) combine for temperature independance.

$$\frac{dV_{BE}}{dT} \approx -2 \text{ mV}/^\circ\text{C}.$$

$$\frac{d(\Delta V_{BE})}{dT} = \frac{k}{q} \ln(k) \approx 0.18 \text{ mV}/^\circ\text{C}$$

Setting the sum to zero around 27°C .

$$\frac{dV_{BG}}{dT} = \cancel{\frac{dV_{BE}}{dT}} + \frac{R_2}{R_1} \frac{d(\Delta V_{BE})}{dT} = 0.$$

$$-2 + \frac{R_2}{R_1} \cdot 0.018 = 0 \Rightarrow \frac{R_2}{R_1} = \frac{2}{0.018} \approx 11.1$$

Choosing Resistor value,

$$R_1 = 1 \text{ k}\Omega.$$

$$R_2 = 11.1 \text{ k}\Omega.$$

Calculating Bandgap Output Voltage:-

$$\Delta V_{BE} = \frac{kT}{q} \ln(k) \approx 0.02585 \times 2.079 = 0.0538 \text{ V}.$$

At the room temperature, $V_{BE} \approx 0.7 \text{ V}$.

$$V_{BG} = 0.7 + 11.1 \times 0.0538 = 0.7 + 0.597 = 1.297 \text{ V}.$$

Current Reference (I_b).

A PTAT current is required,

$$\Delta V_{BE} = \frac{kT}{q} \ln(k) ; I_{PTAT} = \frac{\Delta V_{BE}}{R_1}$$

$$I_b = \frac{\frac{kT}{q} \ln k}{R_1}$$

$$I_b = \frac{0.0538}{1000} = 53.8 \mu\text{A}.$$

Load Resistor, R_L ,

$$R_L \parallel (a_2 \text{ output impedance}) \approx 10 - 50 \text{ k}\Omega.$$

Start-up Circuit,

→ Adding a small NMOS with its gate tied to V_{BG} , drain to the internal bias, (node @ the source of top PMOS in mirror) source to ground.

At startup: V_{BG} is at 0V, so the NMOS is ON and provides current to kickstart the circuit

When V_{BG} rises: Once V_{BG} exceeds threshold, NMOS turns off (leakage only)

$$P_{\text{Startup, on}} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{DD} - V_{th})^2$$